BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, MULTAN

INTERMEDIATE (11TH CLASS) EXAMINATION, 2024

PHYSICS

Past Paper 2024 • Paper Code: 2471 / 2472

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

PHYSICS PAPER-I GROUP-I (Objective Type)

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 2471

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill that bubble in front of that question number, on bubble sheet. Use marker or pen to fill the bubbles. Cutting or filling two or more bubbles will result in zero mark in that question.

Q1. How many colours are used by colour printing to produce the entire range of colours?

(A) 3 (B) 4

(C) 5 (D) 6

Q2. The dimensions of the relation mc^2 are equal to the dimensions of:

(A) Force (B) Momentum

(C) Heat (D) Velocity

Q3. The resultant of two forces $\vec{F}_1$ and $\vec{F}_2$ making an angle of 90° with each other is:

(A) $(F_1 - F_2)^2$ (B) $F_1 + F_2$

(C) $(F_1 + F_2)^2$ (D) $\sqrt{(F_1^2 + F_2^2)}$

Q4. The magnitude of $\hat{j}\cdot(\hat{k}\times\hat{i})$ is equal to:

(A) 1 (B) $2\hat{j}$

(C) 0 (D) $-2\hat{j}$

Q5. The velocity of a body changes with constant rate. The acceleration is:

(A) Zero (B) Negative

(C) Constant (D) Increases

Q6. The velocity time graph of a body is shown. It implies that: [figure]

(A) Force is positive
(B) Force is negative
(C) Force is zero
(D) Force is constant

Q7. Gravity performs zero work when body accelerates:

(A) Vertically upward
(B) Vertically downward
(C) Inclined plane
(D) In a vertical loop

Q8. The acceleration of an object falling freely is:

(A) $9.8ms^{-2}$
(B) $0ms^{-2}$
(C) $-9.8ms^{-2}$
(D) $5ms^{-2}$

Q9. The rotational K.E of any ring of radius 'r' is given by:

(A) $(1/2)r\omega^2$
(B) $(1/2)mr^2\omega^2$
(C) $(1/2)mr^2$
(D) $(1/4)mr^2\omega^2$

Q10. The viscosity of water at 30°C is:

(A) $0.019Nm^{-2}s$
(B) $1000Nm^{-2}s$
(C) $1Nm^{-2}s$
(D) $0.801Nm^{-2}s$

Q11. The time period of a simple pendulum, whose length is 980m is:

(A) 2π sec
(B) $2\pi\sqrt{0.1}$ sec
(C) 20π sec
(D) $2/\pi$ sec

Q12. The speed of sound wave is independent of:

(A) Pressure
(B) Medium
(C) Source of sound
(D) Temperature

Q13. A longitudinal sinusoidal wave has wavelength of 1cm with a time period of 2sec, its wave velocity is:

(A) $50ms^{-1}$
(B) $0.005ms^{-1}$
(C) $0.5ms^{-1}$
(D) $2ms^{-1}$

Q14. Which one of the given cannot be polarized?

(A) Light waves
(B) Radio waves
(C) Microwaves
(D) Sound waves

Q15. The minimum number of rays required by a lens to form an image are:

(A) 2
(B) 3
(C) 4
(D) 5

Q16. When heat is removed from the system, entropy is:

(A) Remain same
(B) Positive
(C) Negative
(D) Zero

Q17. For mono atomic gas $C_v = (3/2)R$, therefore gamma "ɣ" for gas is:

(A) 3/5
(B) 2/5
(C) 2
(D) 5/3

PHYSICS PAPER-I GROUP-I (Subjective Type)

Time Allowed: 2.40 Hours | **Maximum Marks:** 68

Note: Write same question number and its parts number on answer book, as given in the question paper.

Q. SECTION-I

Q2. (Marks 16) Attempt any eight parts.

(i) How do you check the correctness of an equation?

(ii) How would a numerical data should be rounded off up to last significant figure?

(iii) What do you understand about precise and accurate measurement?

(iv) An old saying is that "A chain is only as strong as its weakest link" what analogous statement can you make regarding experimental data used in computation?

(v) Two vectors have unequal magnitude. Can their sum be zero? Explain.

(vi) What is the minimum value of tension in the string? [figure]

(vii) How do you subtract two vectors?

(viii) An object is thrown vertically upward. Discuss the sign of acceleration due to gravity relative to velocity, while the object is in air.

(ix) How a rocket is propelled in space?

(x) When a moving car stops quickly, in what direction passengers fall and why?

(xi) What is the method of fermentation?

(xii) What sort of energy is in (a) compressed spring (b) moving car (c) water in a high dam?

Q3. (Marks 16) Attempt any eight parts.

(i) If a body of mass 10kg is allowed to fall freely what will be its weight?

(ii) Show that orbital angular momentum, $L_o = mvr$.

(iii) What is meant by moment of inertia? Explain its significance.

(iv) Why does a diver change his body position before and after diving in the pool?

(v) Explain the term viscosity.

(vi) Why fog droplets appear to be suspended in air?

(vii) What is second pendulum also write its length, time period and frequency?

(viii) Can we realize an ideal simple pendulum?

(ix) Describe some common phenomena in which resonance plays an important role?

(x) A wave has speed 400 m/sec. Find wavelength of a wave if frequency is 2 kHz.

(xi) Explain why sound travels faster in warm air than in cold air?

(xii) What features do longitudinal waves have in common with transverse waves?

Q4. (Marks 12) Attempt any six parts.

(i) How is the distance between interference fringes affected by the separation between the slits of Young's experiment? Can fringes disappear?

(ii) How interference produced in their film?

(iii) Could you obtain Newton's rings with transmitted light? If yes, would be pattern be different from that obtained with reflected light?

(iv) What is Optical fibre? Write its types.

(v) What is the function of turn table in the spectrometer?

(vi) If a person was looking through a telescope at the full moon, how would the appearance of the moon be changed by covering half of the objective lens?

(vii) State second law of thermodynamics in terms of entropy.

(viii) Can the mechanical energy be converted into heat energy? If so give an example.

(ix) A thermos flask containing milk as a system is shaken rapidly. Does the temperature of the milk rise?

Q. SECTION-II

Q. (Marks 24) NOTE: Attempt any three questions.

Q5.

(a) [5] Explain what is meant by projectile motion? Describe the expression for (i) Height of the projectile (ii) Time of flight

(b) [3] Find the projection of vector $\vec{A} = 2\hat{i} - 8\hat{j} + \hat{k}$ in the direction of vector $\vec{B} = 3\hat{i} - 4\hat{j} - 12\hat{k}$

Q6.

(a) [5] How would you portray step by step guide for interconversion of PE and KE?

(b) [3] Find the temperature at which the velocity of sound in air is two times its velocity at 10°C.

Q7.

(a) [5] Define real and apparent weight and discuss when apparent weight increases, decreases and becomes zero during vertical motion.

(b) [3] An 8.0kg body executes S.H.M with amplitude 30cm. The restoring force is 60N. When the displacement is 30cm. Find (i) period (ii) speed when the displacement is 12cm

Q8.

(a) [5] Bernoulli's equation represents the conservation of energy in fluid dynamics. Discuss it.

(b) [3] Show that the ratio of the root mean square speeds of molecules of two different gases at a certain temperature is equal to the square root of the inverse ratio of their masses.

Q9.

(a) [5] Describe the experiment performed by Michelson to find the speed of light. Also discuss the speed of light reduced in other materials than vacuum.

(b) [3] Light of wavelength 450nm is incident on a diffraction grating, on which 5000 lines per centimeter have been ruled. How many orders of spectra can be observed on either side of the direct beam?

GROUP – II

PHYSICS PAPER-I GROUP-II (Objective Type)

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 2472

> Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill that bubble in front of that question number, on bubble sheet. Use marker or pen to fill the bubbles. Cutting or filling two or more bubbles will result in zero mark in that question.

Q1. Number of significant zeroes in 3.50070 are:

(A) 1
(B) 2
(C) 3
(D) No significant zero

Q2. If we add the numbers 2.7543, 4.10, 1.273, the rounded off answer will be:

(A) 8.1273 (B) 8.127

(C) 8.2 (D) 8.13

Q3. If vector $\vec{A}$ makes an angle θ with Y–axis, then its Y-component will be:

(A) A sin θ (B) A cos θ

(C) A tan θ (D) A

Q4. The magnitude of $\vec{A} = \cos\theta\,\hat{i} + \sin\theta\,\hat{j}$ is:

(A) $\sqrt{(\cos\theta + \sin\theta)}$ (B) 2

(C) 1 (D) $\sqrt{(1+\cos^2\theta)}$

Q5. A body moves in a circle of radius r. The displacement covered in one rotation is:

(A) 2πr (B) πr

(C) (π/2)r (D) Zero

Q6. A stone is dropped from the top of a tower. It takes 2s to reach the ground. The height of the tower is:

(A) 19.6 m (B) 9.8 m

(C) 40 m (D) 19 m

Q7. Two masses M and 4M are moving with same K.E. The ratio of their linear momenta is:

(A) 1 : 16 (B) 1 : 2

(C) √2 : 1 (D) 4 : 1

Q8. Which is a correct relation?

(A) $\vec{v} = \vec{r} \times \vec{w}$ (B) $\vec{w} = \vec{v} \times \vec{r}$

(C) $\vec{v} = \vec{w} \cdot \vec{r}$ (D) $\vec{v} = \vec{w} \times \vec{r}$

Q9. A body of mass m is moving in a vertical circle of radius r, tied with a string. The tension at the lowest point is:

(A) $T = m((v^2/r) - g)$ (B) $T = m((v^2/r) + g)$

(C) $T = m(v^2 + gr)$ (D) $T = m(v^2 + gr)$

Q10. 1 torr = _____ N/m^2

(A) 133.33 (B) 123.33

(C) 122.22 (D) 143.33

Q11. What will be the displacement of a particle in SHM when its velocity is half the maximum velocity (amplitude = x_o):

(A) $(3/\sqrt{2})x_o$ (B) $\sqrt{2}x_o$

(C) $(3/4)x_o$ (D) $(\sqrt{3}/2)x_o$

Q12. A physical system under going forced vibrations in known as:

(A) Simple harmonic oscillator
(B) Driven harmonic oscillator
(C) Damped harmonic oscillator
(D) Torsional oscillator

Q13. The frequency of sound emitted from a source in water is 600 Hz. If speed of sound in water and air is 1500 m/s and 300 m/s respectively, then frequency of sound heard above the water surface is:

(A) 300 Hz
(B) 750 Hz
(C) 600 Hz
(D) 120 Hz

Q14. Which monochromatic light will produce maximum orders of spectra using a diffraction grating?

(A) Blue
(B) Red
(C) Green
(D) Yellow

Q15. Multimode step index fibre is useful for short distance to carry white light due to:

(A) Polarization effects
(B) Diffraction effects
(C) Interference effects
(D) Dispersion effects

Q16. In PV graph of isothermal and adiabatic process, the adiabatic curve has _____ work under the curve, than isothermal curve,:

(A) Greater
(B) Smaller
(C) Equal
(D) Negative work

Q17. A fixed mass of an ideal gas in a cylinder is compressed isothermally. Which is true statement?

(A) Heat is dissipated from the system
(B) K.E. of the gas increases
(C) P.E of the gas increases
(D) No work is done on the gas

PHYSICS PAPER-I GROUP-II (Subjective Type)

Time Allowed: 2.40 Hours | **Maximum Marks:** 68

> Note: Write same question number and its parts number on answer book, as given in the question paper.

Q. SECTION-I

Q2. (Marks 16) Attempt any eight parts.

(i) Differentiate between random error and systematic error.

(ii) What is principle of homogeneity?

(iii) Name several repetitive phenomenon occurring in nature which could serve as reasonable time standards?

(iv) Give the drawbacks to use the period of a pendulum as a time standard?

(v) Why the cross product is not commutative? Explain briefly.

(vi) The vector sum of three vectors gives a zero resultant. What can be the orientation of the vectors?

(vii) If one of the rectangular components of a vector is not zero, can its magnitude be zero? Explain.

(viii) Can the velocity of an object reverse direction when acceleration is constant? If so give an example.

(ix) Calculate the force due to water when it flows out from a pipe at $3kgs^{-1}$ and its velocity changes from $5ms^{-1}$ to zero on striking the wall?

(x) An object is thrown vertically upward. Discuss the sign of acceleration due to gravity, relative to velocity, while the object is in air?

(xi) Derive a relation between power and velocity.

(xii) A person holds a bag of groceries while standing still, talking to a friend. A car is standing still while its engine is running. From stand point of work, how are these two situations similar?

Q3. (Marks 16) Attempt any eight parts.

(i) Show that orbital angular momentum $L_o = mvr$.

(ii) What is meant by moment of inertia? Explain its significance.

(iii) Prove that 1 radian = 57.3°.

(iv) Write down applications of communication satellites.

(v) What are the factors upon which drag force acting upon a small sphere of radius "r" moving through a liquid, depend?

(vi) A chimney works best when it is tall. Why?

(vii) Does frequency depends on amplitude for harmonic oscillators?

(viii) The equation for SHM of an object is given by $X = 0.25\cos(\pi/8)t$. What will be displacement after 2 seconds?

(ix) What is Hook's law? Write its mathematical form.

(x) Explain why sound travels faster in warm air than in cold air.

(xi) How will you differentiate between longitudinal and transverse wave?

(xii) What is period of 300 cycles per second of sound waves?

Q4. (Marks 12) Attempt any six parts.

(i) Why interference is necessary to produce diffraction pattern? Answer this question with the analytical approach.

(ii) Explain the term "Optical rotation".

(iii) Could you obtain Newton's rings with transmitted light? If yes, would the pattern be different from that obtained with reflected light?

(iv) Explain how a convex lens is used as a magnifier?

(v) Explain scattering and absorption as a loss of power?

(vi) If a person was looking through a telescope at the full moon, how would the appearance of the moon be changed by covering half of the object lens?

(vii) What happens to the temperature of room, when an air conditioner is left running on a table in the middle of the room?

(viii) Is it possible to convert internal energy into mechanical energy? Explain with an example.

(ix) Calculate the work done in the given diagram: [figure]

Q. SECTION-II

Q. (Marks 24) NOTE: Attempt any three questions.

Q5.

(a) [5] What do you know about collision? How would two balls collides elastically in different cases?

(b) [3] The magnitude of dot and cross product of two vectors are $6\sqrt{3}$ and "6" respectively. Find the angle between vectors.

Q6.

(a) [5] Define absolute potential energy. Derive its mathematical expression $U = -GMm/r$

(b) [3] An organ pipe has a length of 50cm. Find the frequency of its fundamental note when it is closed at one end. (Speed of sound = 350 ms^{-1}).

Q7.

(a) [5] How orbital radius of Geostationary orbits are calculated mathematically. Also calculate its value and its height from the earth surface.

(b) [3] A spring, whose spring constant is 80.0 Nm^{-1} vertically supports a mass of 1.0 kg in the rest position. Find the distance by which the mass must be pulled down, so that on being released, it may pass the mean position with a velocity of 1.0 ms^{-1}.

Q8.

(a) [5] State and prove the Bernoulli's equation in dynamic fluid that relates pressure to fluid speed and height.

(b) [3] 336J of energy is required to melt 1g of ice at 0°C. What is the change in entropy of 30g of water at 0°C as it is changed to ice at 0°C by a refrigerator.

Q9.

(a) [5] What is compound microscope? Describe its construction and working. Also calculate its magnifying power.

(b) [3] A light is incident normally on a grating which has 2500 lines per centimeter. Compute the wavelength of a spectral line for which the deviation in second order is 15.0°.